

KINETICS OF CURVILINEAR TRANSLATION

Differential Equation of curvilinear motion

According to Newton's second law, when an unbalanced external force acts on a body, it is accelerated in the direction of it. When the direction of the external force changes, the body will be moving along a curved path. Even when the force does not change its direction, if the body is having an initial motion not in line with the force, then also the particle will be undergoing a curvilinear motion.

In kinematics, we have seen that the acceleration of a body undergoing curvilinear translation can be resolved in to two mutually perpendicular directions either along x and y directions or along the normal and tangential to the path. Accordingly, we can write the differential equation of curvilinear motion as

$$\frac{W}{g} \ddot{x} = \Sigma F_x \quad \text{and} \quad \frac{W}{g} \ddot{y} = \Sigma F_y$$

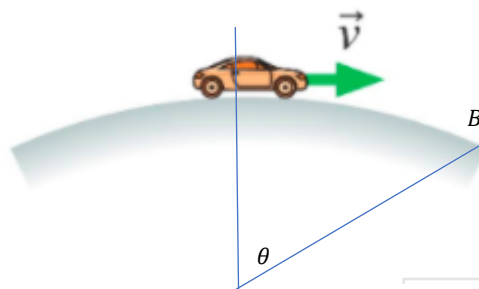
Or

$$\frac{W}{g} a_n = \Sigma F_n \quad \text{and} \quad \frac{W}{g} a_t = \Sigma F_t$$

That is, the curvilinear translation can be considered as two simultaneous rectilinear translations along two mutually perpendicular directions.

#Example 1

A car of total weight $W = 2225 \text{ N}$ travels in a vertical plane with constant speed $v = 72 \text{ km/hr}$ along a circular curve of radius $\rho = 300 \text{ m}$. Find the reaction R exerted on the car by the track as it passes over the crest of the curve. Also find at which point B on the curve as defined by the angle θ , will the road reaction become equal to zero?

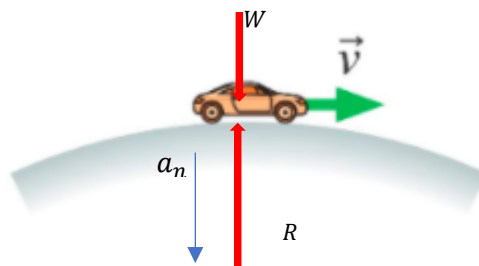


Since the car is moving with constant velocity, its tangential acceleration will be equal to zero.

$$v = 72 \frac{\text{km}}{\text{hr}} = 20 \text{ m/s}$$

$$a_t = \frac{dv}{dt} = 0$$

$$a_n = \frac{v^2}{\rho} = \frac{20^2}{300} = 1.33 \text{ m/s}^2$$



The equation of motion in the normal direction

$$\frac{W}{g} a_n = \Sigma F_n = W - R$$

$$R = 2225 \left(1 - \frac{1.33}{9.81} \right) = 1923.3 \text{ N}$$

Let, at B defined by the angle θ , the road reaction becomes equal to zero.

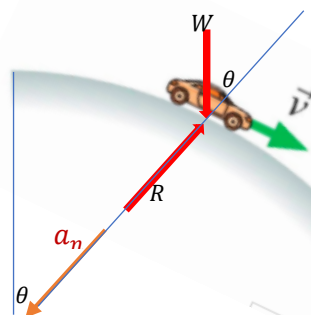
Resolving the forces again in the normal direction to the curve at B,

$$\frac{W}{g} a_n = \Sigma F_n = W \cos \theta - R$$

But $R = 0$

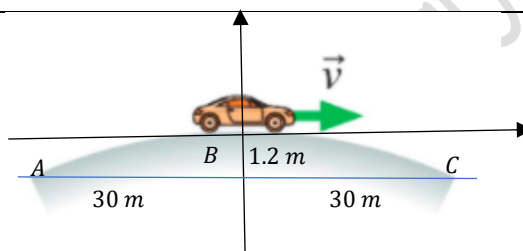
$$\therefore \cos \theta = 0.136$$

$$\theta = 82.2^\circ$$



Example 2

A car of weight W travels with uniform speed v over a vertical parabolic path ABC. Determine the force exerted on the road by the wheels of the car as it passes over the crest B of the curve if $v = 27 \text{ m/s}$



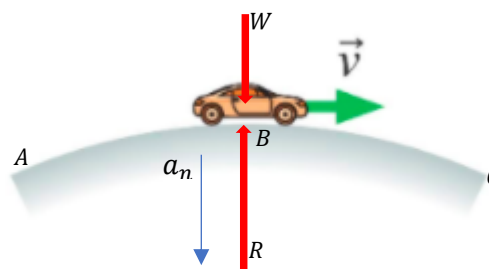
The equation of the parabola can be considered as $y = -kx^2$

When $x = 30 \text{ m}$, $y = -1.2 \text{ m}$

$$-1.2 = -k \cdot 30^2$$

$$k = \frac{1}{750}$$

$$\therefore y = -\frac{1}{750}x^2$$



Since the car is moving with uniform velocity, its tangential acceleration is zero. The normal acceleration, $a_n = \frac{v^2}{\rho}$. To find the normal acceleration, we need to find out the radius of curvature of the parabola at B(0,0).

$$\frac{1}{\rho} = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}$$

$$\frac{dy}{dx} = -\frac{2}{750}x \quad \lll \left(\frac{dy}{dx}\right)_{B(0,0)} = 0 \quad \lll \left(\frac{d^2y}{dx^2}\right)_{B(0,0)} = -\frac{2}{750}$$

$$\frac{1}{\rho} = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}} = -\frac{2}{750} \quad ; \quad \left|\frac{1}{\rho}\right| = \frac{2}{750}$$

$$a_n = \frac{v^2}{\rho} = 27^2 \times \frac{2}{750} = 1.94 \text{ m/s}^2$$

Writing down the equation of motion in the normal direction,

$$\frac{W}{g} a_n = \Sigma F_n = W - R$$

$$R = W \left(1 - \frac{1.94}{9.81}\right) = 0.8 W$$

#Exercise 36

36.1

A ball of weight W attached to a string of length l whirls in a horizontal circular path with a uniform velocity v . Find the tensile force in the string.

36.2

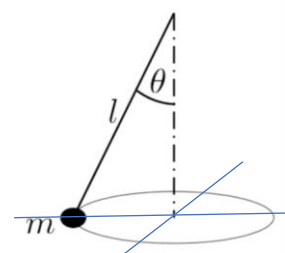
Small oscillations of a simple pendulum of weight W and length l are defined by the equation $s = s_0 \cos pt$, where s_0 is the amplitude of swing and p is a constant equal to $\sqrt{g/l}$. Determine the maximum value of tension in the string.

36.3

The coefficient of friction between wet asphalt pavement and tyres of an automobile is $\mu = 0.2$. At what uniform speed v can the automobile travel round a curve of radius $\rho = 250 \text{ m}$ without skidding sideways if the road is level?

36.3

The bob of a conical pendulum of length l and weight W describes a horizontal circle defined by equations $x = a \cos \omega t$ and $y = a \sin \omega t$, where a is the radius of circular path and ω is a constant. Find the tension in the string.



Prof. Biju N., School of Engineering, CUSAT